INF273 – Assignment #4

Use the following adjusted Simulated Annealing and define your own operators!

You should define 3 operators!

You should demonstrate creativity in designing these new operators!

Problem-dependent operators are highly recommended! These operators utilize input data and take advantage of knowledge about the problem.

You should briefly explain why you expect the proposed operator to be beneficial in supporting your approach.

You should implement two versions of the algorithm. One with equal weights for all three operators during the search (P1 = P2 = P3), and the second is to run it with tuned weights (through trial and error).

You should fill in the table below. Ensure that all previous results from assignments 2 and 3 are retained in the table.

Report the best solutions obtained from the simulated annealing using the new operators (selecting the best of the two new implementations).

Similar to assignment #3, run each of the 8 instances 10 times. (Report the time in seconds)

The quality and robustness of the operators are more important than their runtime.

Although there is no limit and you are completely free to design your operator, we reserve the right to reject an idea and request a resubmission. (This has never happened before!)

You must submit a PDF file containing the tables and best solutions of this new implementation at the beginning, followed by a brief explanation (support) of the operators. Ensure that the solutions can be selected and copied from the PDF file. Do not include screenshots in the PDF.

This time, you must also submit your code (in a zip file) separately, labeled as "Assignment #4, Codes"

Instance name (e.g. R_20)				
	Average objective	Best objective	Improvement (%)	Running time in seconds
Random Search	Copy from A#2	Copy from A#2	Copy from A#2	Copy from A#2
Local Search-1-insert	Copy from A#3	Copy from A#3	Copy from A#3	Copy from A#3
Simulated Annealing-1-insert	Copy from A#3	Copy from A#3	Copy from A#3	Copy from A#3
SA-new operators (equal weights)				
SA-new operators (tuned weights)				

Simulated Annealing (modified for assignment #4)

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Input: initial solution (s_0),
Input: neighborhood operators: OP1, OP2, and OP3
Parameters: T_f (final temperature) = 0.1
Parameters: P_1 (probability of using OP1), P_2, and P_3
Input: evaluation function f, f(s) ⟶ the cost of s
Incumbent ⇐ s_0 , BestSolution ⇐ s_0
for w = 1 to 100
    NewSolution ⇐ select & apply an Operator on Incumbent (Roulette wheel selection)
    ΔE ⇐ f(NewSolution) − f(Incumbent)
    if NewSolution is feasible and ΔE < 0 then
        Incumbent ⇐ NewSolution
        if f(Incumbent) < f(BestSolution) then
            BestSolution ⇐ Incumbent
        end if
    elseif NewSolution is feasible
        if Rand < 0.8 then
            Incumbent ⇐ NewSolution
        end if
        Δ_w ⇐ ΔE
    end if
end for
DeltaAvg = mean (Δ_w)
T_0 = −DeltaAvg / ln(0.8) ,  α = ⁹⁹⁰⁰√(T_f / T_0)
T ⇐ T_0
for iteration = 1 to 9900
    NewSolution ⇐ select & apply an Operator on Incumbent (Roulette wheel selection)
    ΔE ⇐ f(NewSolution) − f(Incumbent)
    if NewSolution is feasible and ΔE < 0 then
        Incumbent ⇐ NewSolution
        if f(Incumbent) < f(BestSolution) then
            BestSolution ⇐ Incumbent
        end if
    elseif NewSolution is feasible and Rand < p = e^(−ΔE/T)
        Incumbent ⇐ NewSolution
    end if
    T = α * T
end for
```